

AP Precalculus Unit 2 - Exponential and Logarithmic Functions

Complete Practice Workbook

15 worksheets - 270 distinct assessed questions

Purpose

This volume combines the fully rewritten practice worksheets for Topics 2.1-2.15. Every topic includes 12 constructed-response questions, 4 AP-style multiple-choice questions, and 2 six-point free-response tasks.

Resource design

- Original, non-repeating questions with a deliberate progression from foundational structure to modeling and justification.
- Explicit calculator guidance, exact-value expectations, domain restrictions, parameter interpretation, and error analysis.
- AP-style multiple-choice and free-response tasks integrated into every topic.
- Consistent ECHS formatting for classroom printing, annotation, and digital distribution.

Topics included

Topic	Title
2.1	Change in Arithmetic and Geometric Sequences
2.2	Change in Linear and Exponential Functions
2.3	Exponential Functions
2.4	Exponential Function Manipulation
2.5	Exponential Function Context and Data Modeling
2.6	Competing Function Model Validation
2.7	Composition of Functions
2.8	Inverse Functions
2.9	Logarithmic Expressions
2.10	Inverses of Exponential Functions
2.11	Logarithmic Functions
2.12	Logarithmic Function Manipulation
2.13	Exponential and Logarithmic Equations and Inequalities
2.14	Logarithmic Function Context and Data Modeling
2.15	Semi-log Plots



ECHS Mathematics
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Topic 2.1 - Change in Arithmetic and Geometric Sequences

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Describe arithmetic change as repeated addition and geometric change as repeated multiplication.
- Move among recursive, explicit, numerical, graphical, and contextual representations.
- Use discrete domains, precise notation, and contextual units.

Section A: Foundation and representation

Question 1

Core skill**No calculator**

Determine whether each sequence is arithmetic, geometric, or neither. Justify with differences or ratios: (a) 7, 12, 17, 22, ...; (b) 81, 27, 9, 3, ...; (c) 2, 6, 12, 20, ...

Question 2

Core skill**No calculator**

Write both a recursive and an explicit rule for $-9, -2, 5, 12, \dots$, using $a_1 = -9$.

Question 3

Core skill

No calculator

A geometric sequence begins $160, -80, 40, -20, \dots$. Find g_8 .

Question 4

Core skill

No calculator

An arithmetic sequence has $a_5 = -11$ and $a_{18} = 54$. Find a_{30} .

Question 5

Core skill

No calculator

A positive geometric sequence has $g_3 = 50$ and $g_7 = 4050$. Find g_1 and g_n .

Question 6

Core skill

No calculator

The table gives a sequence. Fill in the missing values and identify the invariant.

n	0	1	2	3	4
p_n	96	72	54	$\square$	$\square$

Section B: Application, comparison, and error analysis

Question 7

Modeling

No calculator

A theater has 24 seats in row 1 and each successive row has 3 more seats. Write a rule for row n , find the seats in row 18, and state the appropriate domain if the theater has 25 rows.

Question 8

Modeling

Calculator permitted

A culture contains 240 cells at hour 0 and retains 82% of its cells each hour. Write the model and determine when the population first falls below 100 cells.

Question 9

Comparison

Calculator permitted

Compare the two rules $A_n = 40 + 18n$ and $B_n = 40(1.18)^n$. Find the smallest whole-number n for which $B_n > A_n$.

Question 10

Error analysis

No calculator

A student says, "Because the common ratio is 0.72, the sequence decreases by 72% each step." Diagnose the error and write the correct percent change.

Question 11

Core skill

No calculator

Construct an arithmetic sequence whose fifth term is 9 and whose twelfth term is -12 . Then determine whether the twentieth term is positive, negative, or zero.

Question 12

Core skill

No calculator

Create a geometric sequence with $h_1 = 6$ and $h_4 = -48$. State whether the sequence alternates in sign and explain why.

Section C: AP-style multiple choice**MCQ 1**

No calculator

Which rule represents an arithmetic sequence with $a_4 = 17$ and $a_9 = -3$?

- A. $a_n = 29 - 4n$
- B. $a_n = 17 - 4(n - 1)$
- C. $a_n = 5(-4)^{n-1}$
- D. $a_n = -3 + 4(n - 9)$

MCQ 2

No calculator

A sequence is defined by $g_0 = 14$ and $g_n = 1.2g_{n-1}$. Which statement is correct?

- A. The sequence increases by 1.2 each term.
- B. The sequence increases by 20% each term.
- C. The sequence decreases by 80% each term.
- D. The sequence is arithmetic with difference 0.2.

MCQ 3

No calculator

Which observation is sufficient to prove that a sequence is geometric?

- A. The first four terms look curved when plotted.
- B. Three consecutive differences are equal.
- C. A stated rule satisfies $a_{n+1} = ra_n$ for every allowed n .
- D. Four consecutive ratios happen to be close.

MCQ 4

No calculator

The graph of a sequence is shown as isolated points. Why should the points normally remain unconnected?

- A. Connected graphs are never used in algebra.
- B. The outputs between integer indices are all zero.
- C. The sequence domain is discrete unless a continuous extension is defined.
- D. Connecting points changes the common difference.

Section D: Free-response reasoning

FRQ 1 - Coordinating representations

Calculator permitted

6 points

At week 0, Club A has 100 QAR and adds 40 QAR each week. Club B has 100 QAR and increases by 25% each week. (a) Write explicit rules. (b) Make a table for weeks 0 through 4 and identify the invariant for each sequence. (c) Compare the balances at week 8. (d) State a reasonable domain and one modeling assumption.

FRQ 2 - Evidence and limitations

Calculator permitted

6 points

A sequence has values 192, 144, 108, 81 at indices 0, 1, 2, 3. (a) Show evidence for a plausible family. (b) Write recursive and explicit rules for that model. (c) Predict the value at index 7. (d) Explain why the four values support but do not prove the continuation.

Topic 2.2 - Change in Linear and Exponential Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Distinguish additive change from proportional change across tables, graphs, formulas, and contexts.
- Construct linear and exponential functions from two values or from stated rates.
- Compare models, interpret intersections, and respect discrete versus continuous domains.

Section A: Foundation and representation

Question 1

Core skill

No calculator

For each table, decide whether the data are exactly linear, exactly exponential, or neither.

x	0	1	2	3
$A(x)$	14	20	26	32
$B(x)$	14	21	31.5	47.25
$C(x)$	14	19	27	38

Question 2

Core skill

No calculator

Construct the linear function through $(-2, 11)$ and $(5, -10)$.

Question 3

Core skill

No calculator

Construct an exponential function $f(x) = ab^x$ through $(0, 6)$ and $(3, 48)$.

Question 4

Core skill

No calculator

A quantity increases from 250 to 310 in one year. Write both (i) the additive change and (ii) the multiplicative factor and percent change.

Question 5

Core skill

No calculator

The function $P(t) = 900(0.94)^t$ models a quantity after t days. State the initial value, daily percent change, and whether the graph is increasing or decreasing.

Question 6

Equivalent intervals

No calculator

A linear model changes by 18 units every 3 hours. An exponential model changes by a factor of 1.331 every 3 hours. Express each change per hour.

Section B: Application, comparison, and error analysis**Question 7**

Comparison

Calculator permitted

At $x = 0$, a linear model and an exponential model both equal 50. The linear model adds 12 per unit; the exponential model grows by 16% per unit. Build both functions and determine which is larger at $x = 5$.

Question 8

Comparison

Calculator permitted

For $L(x) = 30 + 20x$ and $E(x) = 30(1.25)^x$, find the first whole-number x for which $E(x) > L(x)$.

Question 9

Error analysis

No calculator

A student sees a curve bending upward and concludes that the function is exponential. Give two reasons that the conclusion is not justified.

Question 10

Modeling

No calculator

A water tank contains 680 L and drains at 35 L per minute. Another tank contains 680 L and loses 7% of its contents per minute. Write both models and state a physically reasonable domain for each.

Question 11

Core skill

No calculator

Find the exponential function whose values at $x = 2$ and $x = 5$ are 40 and 135, respectively. Leave the growth factor in exact form.

Question 12

Construction

No calculator

Create one linear and one exponential function that both pass through $(0, 24)$ and $(2, 54)$. Compare their values at $x = 4$.

Section C: AP-style multiple choice

MCQ 1

No calculator

Which table represents exponential growth?

- A. y : 4, 9, 14, 19
- B. y : 4, 8, 16, 32
- C. y : 4, 7, 12, 19
- D. y : 4, 2, 0, -2

MCQ 2

No calculator

A function increases by 30% whenever x increases by 2. What is the factor for a one-unit increase in x ?

- A. 1.15
- B. 1.30
- C. $\sqrt{1.30}$
- D. 0.70

MCQ 3

No calculator

Which statement correctly compares $L(x) = 200 + 15x$ and $E(x) = 200(0.97)^x$ for $x \geq 0$?

- A. Both increase.
- B. L increases while E decreases.
- C. Both decrease.
- D. E eventually exceeds L .

MCQ 4

No calculator

Two models intersect at $(6, 410)$. Which interpretation is valid?

- A. Both models are correct for all inputs.
- B. At input 6, both models predict output 410.
- C. The real quantity must equal 410.
- D. The functions have the same rate at $x = 6$.

Section D: Free-response reasoning

FRQ 1 - Build and compare two models

Calculator permitted

6 points

A delivery service offers two pricing plans. Plan L charges 90 QAR plus 14 QAR per delivery. Plan E charges 90 QAR and multiplies the total charge by 1.12 for each delivery. (a) Write both models. (b) Compare the predicted charges for 5 deliveries. (c) Determine the first whole-number delivery count for which Plan E exceeds Plan L. (d) Discuss one limitation of each model.

FRQ 2 - Same data, different structures

Calculator permitted

6 points

Two functions pass through $(0, 30)$ and $(4, 150)$. One is linear and one is exponential. (a) Construct both functions. (b) Find each value at $x = 8$. (c) Explain why the functions agree at two inputs but produce very different predictions later. (d) Give a context in which one model would be more defensible than the other.

Topic 2.3 - Exponential Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Analyze the structure, domain, range, monotonicity, concavity, and end behavior of exponential functions.
- Connect equal-interval proportional change to the parameters of an exponential formula.
- Describe and construct transformed exponential functions using precise asymptotic language.

Section A: Foundation and representation

Question 1

Core skill

No calculator

For $f(x) = 3(2)^x$, state the domain, range, y -intercept, horizontal asymptote, monotonicity, concavity, and both end behaviors.

Question 2

Core skill

No calculator

For $g(x) = -4(1/3)^{x-2} + 5$, state the horizontal asymptote, anchor point, range, and whether the function is increasing or decreasing.

Question 3

Core skill

No calculator

Find the y -intercept and exact x -intercept of $h(x) = 6(2)^{x+1} - 9$.

Question 4

Core skill

No calculator

An exponential function has asymptote $y = -3$, range $(-3, \infty)$, and anchor point $(2, 5)$. Write one possible function with base 4.

Question 5

Core skill

No calculator

Construct $f(x) = ab^x + 2$ through $(0, 8)$ and $(2, 56)$.

Question 6

Structure

No calculator

Construct an exponential function with asymptote $y = 10$ that passes through $(1, 6)$ and $(3, 9)$. Explain why the base represents decay even though the function increases.

Section B: Application, comparison, and error analysis

Question 7

Core skill

No calculator

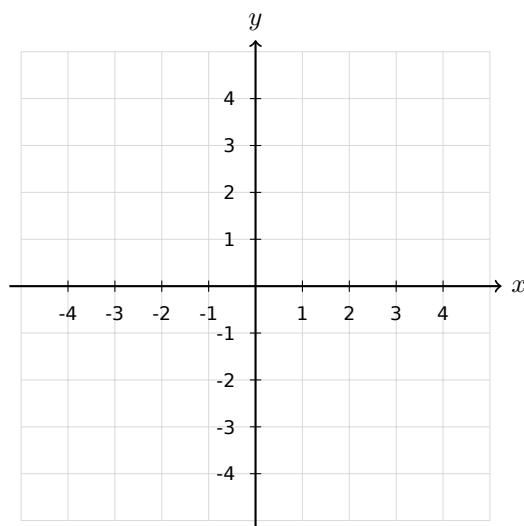
Order the functions 2^x , 3^x , and $(1.4)^x$ from least to greatest for (a) $x > 0$ and (b) $x < 0$.

Question 8

Graphing

No calculator

Sketch $p(x) = 2(1/2)^{x-1} - 4$. Label the asymptote, anchor point, and intercepts.



Question 9

Error analysis

No calculator

A student states that the range of $q(x) = -5(3)^x + 7$ is $(-\infty, \infty)$ because exponentials have all real outputs. Correct the statement.

Question 10

Reasoning

No calculator

Determine whether $f(x) = 4(0.7)^x - 1$ ever equals its horizontal asymptote. Justify algebraically.

Question 11

Modeling

No calculator

The function $T(t) = 18 + 62(0.76)^t$ models a temperature in degrees Celsius. Interpret 18, 62, and 0.76, and describe the long-run behavior.

Question 12

Parameter reasoning

No calculator

Find all values of k for which $f(x) = k(2)^x + 5$ is increasing and has range $(5, \infty)$.

Section C: AP-style multiple choice**MCQ 1**

No calculator

Which function has horizontal asymptote $y = -2$ and range $(-2, \infty)$?

- A. $-3(2)^x - 2$
- B. $3(2)^x - 2$
- C. $3(2)^x + 2$
- D. $-3(1/2)^x + 2$

MCQ 2

No calculator

For $f(x) = a(0.4)^x + 7$, which condition makes f increasing?

- A. $a > 0$
- B. $a < 0$
- C. $a = 0$
- D. No value of a

MCQ 3

No calculator

Which point must lie on $g(x) = 5(3)^{x-4} - 1$?

- A. $(4, 4)$
- B. $(4, 5)$
- C. $(-4, 4)$
- D. $(0, 4)$

MCQ 4

No calculator

An exponential graph approaches $y = 6$ from below as $x \rightarrow -\infty$ and falls without bound as $x \rightarrow \infty$. Which structure is consistent?

- A. $a > 0, b > 1, k = 6$
- B. $a < 0, b > 1, k = 6$
- C. $a > 0, 0 < b < 1, k = 6$
- D. $a < 0, 0 < b < 1, k = 0$

Section D: Free-response reasoning

FRQ 1 - Full feature analysis

No calculator

6 points

Let $f(x) = -6(2)^{x-3} + 10$. (a) State the domain, range, horizontal asymptote, and anchor point. (b) Find the y -intercept and the exact x -intercept. (c) Describe monotonicity, concavity, and end behavior. (d) Explain why the graph cannot cross its asymptote.

FRQ 2 - Construct from transformed data

No calculator

6 points

An exponential function g has horizontal asymptote $y = 4$ and passes through $(1, 10)$ and $(4, 166)$. (a) Determine $g(x)$ in the form $ab^x + 4$. (b) Find $g(0)$ and interpret the point relative to the asymptote. (c) Solve $g(x) = 58$. (d) State the range and describe the effect of replacing a by $-a$.

Topic 2.4 - Exponential Function Manipulation

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Rewrite exponential expressions using exponent properties and common bases.
- Translate growth factors across equivalent input intervals using rational exponents.
- Explain how algebraic equivalence corresponds to shifts, stretches, and unchanged graphs.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Simplify to the form $A(3)^x$: $18(3)^{x-4}$.

Question 2

Core skill

No calculator

Rewrite $6(16)^{x+1}$ using base 2 with a single exponent.

Question 3

Core skill

No calculator

Show that $4(9)^{x/2}$ and $4(3)^x$ are equivalent for all real x .

Question 4

Core skill

No calculator

Rewrite $\frac{5^{2x+3}}{25^{x-1}}$ as a constant.

Question 5

Core skill

No calculator

Find A and c so that $12(8)^{x-2} = A(c)^x$ with c an integer and no horizontal shift.

Question 6

Equivalent intervals

No calculator

A quantity doubles every 9 hours. Write equivalent models using (a) hours and (b) three-hour intervals.

Section B: Application, comparison, and error analysis

Question 7

Equivalent intervals

Calculator permitted

A yearly decay factor is 0.78. Determine the equivalent monthly factor and monthly percent decrease.

Question 8

Modeling

Calculator permitted

An investment grows by 1.5% per quarter. Find the effective annual growth factor and annual percent increase.

Question 9

Core skill

No calculator

Rewrite $f(x) = 10(2)^{x+4} - 7$ in the form $A(2)^x - 7$, and identify the anchor point in the shifted form.

Question 10

Error analysis

No calculator

A student rewrites $(3^x)^2$ as 3^{x+2} . Correct the error and give a numerical check at $x = 1$.

Question 11

Identity

No calculator

Find all real p such that $4(2)^{px} = 4(8)^x$ for every real x .

Question 12

Interpretation

Calculator permitted

Two forms of a model are $P(t) = 900(1.06)^t$ and $P(t) = 900(1.7908477)^{t/10}$. Explain why they are approximately equivalent and what each form emphasizes.

Section C: AP-style multiple choice**MCQ 1**

No calculator

Which expression is equivalent to $7(5)^{x-2}$?

- A. $175(5)^x$
- B. $\frac{7}{25}(5)^x$
- C. $7(5)^x - 2$
- D. $\frac{7}{5}(5)^x$

MCQ 2

No calculator

A factor of 1.21 applies every two months. What is the monthly factor?

- A. 1.105
- B. 1.10
- C. 1.21^2
- D. 0.11

MCQ 3

No calculator

Which equality is valid for every real x ?

- A. $2^{x+4} = 2^x + 16$
- B. $2^{4x} = 8^x$
- C. $9^{x/2} = 3^x$
- D. $(5^x)^3 = 5^{x+3}$

MCQ 4

No calculator

The form $A(4)^x + 3$ is equivalent to $12(4)^{x-1} + 3$. What is A ?

- A. 48
- B. 12
- C. 4
- D. 3

Section D: Free-response reasoning**FRQ 1 - Equivalent forms and interpretation**

Calculator permitted

6 points

A model is written as $P(t) = 2400(1.08)^{t/3}$, where t is measured in months. (a) Identify the factor for a three-month interval. (b) Rewrite the model using a one-month factor. (c) Find the effective annual factor and annual percent increase. (d) Explain what information is clearer in each form.

FRQ 2 - Algebra and graph structure

No calculator

6 points

Consider $f(x) = 18(3)^{x-2} - 5$. (a) Rewrite f in the form $A(3)^x - 5$. (b) Give the anchor point and horizontal asymptote from the original form. (c) Explain why the two forms have identical graphs. (d) Find $f(0)$ and solve $f(x) = 157$ exactly.

Topic 2.5 - Exponential Function Context and Data Modeling

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Construct exponential models from percent rates, two points, transformations, or regression output.
- Use exponential models to predict outputs and solve for inputs.
- Interpret parameters, equivalent time scales, domains, units, and model limitations.

Section A: Foundation and representation

Question 1

Modeling

Calculator permitted

A fund begins with 8500 QAR and grows by 4.2% annually. Write a model and find the balance after 8 years.

Question 2

Modeling

Calculator permitted

A medication amount is 240 mg initially and decreases by 18% each hour. Find the amount after 6 hours and the first whole hour when it is below 50 mg.

Question 3

Core skill

No calculator

A population is 1800 at year 2 and 4050 at year 4. Construct an exponential model in the form $P(t) = ab^t$.

Question 4

Equivalent intervals

Calculator permitted

A quantity triples every 11 days. Write a model with initial value Q_0 , then find the factor for one day.

Question 5

Characteristic time

Calculator permitted

A radioactive sample has mass 96 g and half-life 14 years. Find the mass after 35 years.

Question 6

Interpretation

No calculator

A machine's efficiency is modeled by $E(t) = 72 + 18(0.65)^t$. Interpret all three numerical parameters and state the long-run efficiency.

Section B: Application, comparison, and error analysis**Question 7**

Shifted model

No calculator

A temperature approaches 24°C and satisfies $T(0) = 88$ and $T(4) = 50.2144$. Construct the model.

Question 8

Solve for input

Calculator permitted

Use $P(t) = 1500(1.12)^t$ to solve for the time when $P = 3000$. Give an exact logarithmic expression and a decimal approximation.

Question 9

Error analysis

No calculator

A student fits an exponential regression to six data points and reports $R^2 = 0.998$. Explain why this does not prove that the underlying process is exponential.

Question 10

Equivalent forms

Calculator permitted

Two models for the same data are $A(t) = 400(1.09)^t$ and $B(t) = 400(1.6771)^{t/6}$. Determine whether they are approximately equivalent.

Question 11

Model limitation

No calculator

A town population model predicts $P(20) = 180000$, but the physical capacity is estimated at 120000. What does this discrepancy say about the model?

Question 12

Comparison

Calculator permitted

Construct two different exponential models that both have value 50 at $t = 0$ but one grows by 12% per year and the other doubles every 6 years. Compare their values at $t = 6$.

Section C: AP-style multiple choice**MCQ 1**

No calculator

A quantity decreases by 7% per month. Which factor belongs in an exponential model?

- A. 1.07
- B. 0.93
- C. 0.07
- D. -0.07

MCQ 2

No calculator

A model doubles every 4 hours. Which formula is correct?

- A. $P(t) = P_0 2^{4t}$
- B. $P(t) = P_0 4^{t/2}$
- C. $P(t) = P_0 2^{t/4}$
- D. $P(t) = P_0 (1/2)^{t/4}$

MCQ 3

No calculator

For $T(t) = 18 + 70(0.8)^t$, what does 70 represent?

- A. The limiting temperature
- B. The initial temperature
- C. The initial difference from the limit
- D. The percent decrease

MCQ 4

No calculator

Which statement is the strongest reason not to extrapolate an exponential population model indefinitely?

- A. Exponential functions are never used for populations.
- B. The graph has a horizontal asymptote.
- C. Resource limits can invalidate the constant proportional-growth assumption.
- D. All populations eventually become linear.

Section D: Free-response reasoning**FRQ 1 - Exponential growth model**

Calculator permitted

6 points

A research colony contains 750 organisms at day 0 and 1125 organisms at day 3. (a) Construct an exponential model. (b) Interpret the three-day and one-day growth factors. (c) Predict the population at day 10. (d) Determine when the model first predicts more than 4000 organisms and discuss one limitation.

FRQ 2 - Shifted decay model

Calculator permitted

6 points

A liquid cools toward 22°C . Its temperature is 94°C at time 0 and 46.576°C at time 3. (a) Construct a model $T(t) = 22 + Ab^t$. (b) Interpret A and b . (c) Find the temperature at time 7. (d) Solve for the time when the temperature is 30°C and comment on whether the model ever reaches 22°C .

Topic 2.6 - Competing Function Model Validation

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Select plausible linear, quadratic, or exponential models from patterns, context, and residual behavior.
- Compute and interpret signed residuals and absolute error measures.
- Validate a model within a domain and articulate consequences of overprediction, underprediction, and extrapolation.

Section A: Foundation and representation

Question 1

Core skill**No calculator**

Classify each exact pattern from consecutive outputs: (a) 7, 12, 17, 22; (b) 3, 8, 15, 24; (c) 5, 10, 20, 40.

Question 2

Core skill**No calculator**

For observed $y = 42$ and predicted $\hat{y} = 38.7$, compute the residual and state whether the model overpredicts or underpredicts.

Question 3

Core skill

No calculator

Observed values are 12, 18, 25, 33 and Model A predicts 13, 17, 27, 30. Find all residuals and the MAE.

Question 4

Core skill

No calculator

Model B predicts 11, 20, 24, 34 for the same observed values 12, 18, 25, 33. Compare its MAE with Model A from the previous question.

Question 5

Residual analysis

Calculator permitted

The table shows observed values and predictions.

x	0	1	2	3	4
y	80	64	51	41	33
$\hat{y}_E$	80	64	51.2	40.96	32.768
$\hat{y}_L$	80	68	56	44	32

Compute residuals for both models and select the stronger model.

Question 6

Residual pattern

No calculator

A residual plot for a model shows points steadily rising from about -12 to $+15$ as x increases. What does the pattern indicate?

Section B: Application, comparison, and error analysis**Question 7**

Validation

No calculator

A model with the smallest training error performs poorly on new data. Name two likely explanations.

Question 8

Consequences

No calculator

A hospital demand model has residuals mostly positive during peak hours. Explain the practical risk.

Question 9

Model structure

No calculator

Data for a cooling object are positive and approach room temperature, but an unshifted exponential regression approaches 0. Explain why a shifted exponential model may be more defensible.

Question 10

Context

No calculator

A quadratic model and an exponential model have nearly equal MAE on $0 \leq x \leq 6$. The context is compound interest. Which model is more defensible and why?

Question 11

Extrapolation

No calculator

A regression is fitted using data for ages 10–18. A report uses it to predict age 45. Evaluate the prediction.

Question 12

Model design

No calculator

Construct a residual-based comparison rule that gives extra weight to underprediction when shortages are costly.

Section C: AP-style multiple choice**MCQ 1**

No calculator

Using $r = y - \hat{y}$, a residual of -6 means:

- A. The model underpredicts by 6.
- B. The model overpredicts by 6.
- C. The observed value is negative.
- D. The model is invalid.

MCQ 2

No calculator

Which residual plot most supports a model?

- A. Residuals forming a clear U-shape
- B. Residuals increasing steadily
- C. Residuals scattered around zero with no clear pattern
- D. All residuals positive

MCQ 3

No calculator

Constant second differences are evidence for which exact model family?

- A. Linear
- B. Quadratic
- C. Exponential
- D. Logarithmic

MCQ 4

No calculator

A model has lower MAE but contradicts the known physical mechanism. What is the best conclusion?

- A. Always choose the lower MAE.
- B. Discard context and keep the regression.
- C. Consider both fit and mechanism; the lower MAE alone is not decisive.
- D. Average the two models.

Section D: Free-response reasoning

FRQ 1 - Competing models and residuals

Calculator permitted

6 points

The observed outputs for $x = 0, 1, 2, 3, 4$ are 50, 61, 74, 90, 109. Model L predicts 50, 64, 78, 92, 106. Model E predicts 50, 60, 72, 86.4, 103.68. (a) Compute the residuals for both models. (b) Compute each MAE. (c) Select the better model on this interval and justify using both size and pattern. (d) State one reason the conclusion may change with additional data.

FRQ 2 - Validation in context

Calculator permitted

6 points

A school uses $D(t) = 1200(1.06)^t$ to predict device demand t years after 2026. Actual demand for $t = 0, 1, 2, 3$ is 1200, 1265, 1348, 1460. (a) Compute predictions and residuals. (b) Describe the residual pattern. (c) Evaluate the model for short-term planning. (d) Explain why using the model at $t = 20$ could be risky and name one alternative validation step.

Topic 2.7 - Composition of Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Evaluate compositions from formulas, tables, graphs, and contexts.
- Determine composition domains and explain why order matters.
- Construct and decompose composite functions while interpreting units and intermediate quantities.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Let $f(x) = 3x - 2$ and $g(x) = x^2 + 4$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Question 2

Core skill

No calculator

Let $p(x) = \frac{1}{x-5}$ and $q(x) = 2x + 1$. Find $(p \circ q)(x)$ and its domain.

Question 3

Core skill

No calculator

Let $f(u) = \sqrt{u+6}$ and $g(x) = x^2 - 10$. Find $(f \circ g)(x)$ and its domain.

Question 4

Core skill

No calculator

The tables define f and g .

x	-2	-1	0	1	2
$f(x)$	2	0	1	-1	-2
$g(x)$	1	2	-1	0	2

Find $(f \circ g)(-2)$, $(g \circ f)(-2)$, and $(f \circ g)(1)$.

Question 5

Decomposition

No calculator

Decompose $H(x) = (5x - 1)^4 + 7$ as $f(g(x))$ in two different ways.

Question 6

Core skill

No calculator

Find all x for which $(f \circ g)(x) = 19$ if $f(u) = 2u + 3$ and $g(x) = x^2 - 1$.

Section B: Application, comparison, and error analysis**Question 7**

Context

No calculator

A printer scales a drawing by factor 1.4 and then adds a 2 cm border to each dimension. Let $S(x) = 1.4x$ and $B(x) = x + 4$ for the total dimension after borders on both sides. Write the composite model for an original dimension x .

Question 8

Order and context

No calculator

Compare $B(S(x))$ with $S(B(x))$ from the previous question and interpret the difference.

Question 9

Error analysis

No calculator

A student writes $f(g(x)) = f(x)g(x)$. Give a counterexample using $f(x) = x + 1$ and $g(x) = 2x$.

Question 10

Domain

No calculator

Find the domain of $\ln(4 - x^2)$.

Question 11

Graph structure

No calculator

Let $f(x) = 2^x$ and $g(x) = x - 3$. Describe how $f \circ g$ transforms the graph of f , and compare with $g \circ f$.

Question 12

Construction

No calculator

Construct functions f and g so that $(f \circ g)(x) = \frac{1}{(x+2)^2 + 5}$, with g polynomial and f rational.

Section C: AP-style multiple choice**MCQ 1**

No calculator

If $f(x) = x^2 - 1$ and $g(x) = x + 3$, what is $(f \circ g)(x)$?

- A. $x^2 + 2$
- B. $x^2 + 6x + 8$
- C. $x^2 + 3$
- D. $(x^2 - 1)(x + 3)$

MCQ 2

No calculator

Which restriction belongs to the domain of $\sqrt{5 - 2x}$?

- A. $x \geq 5/2$
- B. $x \leq 5/2$
- C. $x \neq 5/2$
- D. All real x

MCQ 3

No calculator

Which statement is always true?

- A. $f \circ g = g \circ f$
- B. $f(g(x)) = f(x)g(x)$
- C. The output of the inner function must be in the outer function's domain.
- D. Every function can be decomposed uniquely.

MCQ 4

No calculator

If $f(g(4)) = 9$ and $g(4) = 2$, which statement follows?

- A. $f(4) = 9$
- B. $f(2) = 9$
- C. $g(9) = 4$
- D. $g(2) = 9$

Section D: Free-response reasoning**FRQ 1 - Formula, domain, and order**

No calculator

6 points

Let $f(u) = \sqrt{u-2}$ and $g(x) = 6 - x^2$. (a) Find $(f \circ g)(x)$ and its domain. (b) Find $(g \circ f)(x)$ and its domain. (c) Evaluate both compositions at $x = 2$ when defined. (d) Explain why the domains and formulas differ.

FRQ 2 - Composite process in context

Calculator permitted

6 points

A laboratory converts a raw sensor reading x to concentration by $C(x) = 0.8x - 1.5$ mg/L. A reporting system then converts concentration c to a risk score by $R(c) = 12 \ln(c + 1)$. (a) Write the risk score as a function of the raw reading. (b) Determine its domain. (c) Find the score for $x = 5$. (d) Explain why reversing the order is not meaningful in this context.

Topic 2.8 - Inverse Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Determine whether a function is invertible on a stated domain.
- Construct and verify inverse functions algebraically, numerically, and graphically.
- Interpret inverse inputs, outputs, domains, ranges, and restrictions in context.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Determine whether the relation $\{(-2, 5), (0, 1), (3, 5), (7, -4)\}$ defines an invertible function. Justify.

Question 2

Core skill

No calculator

Find the inverse of $f(x) = 4x + 11$ and verify one composition.

Question 3

Core skill

No calculator

Find the inverse of $g(x) = \frac{x-6}{3}$.

Question 4

Core skill

No calculator

Find the inverse of $h(x) = x^3 - 8$ and state both domains.

Question 5

Core skill

No calculator

Restrict the domain of $p(x) = x^2 - 6x + 13$ to $x \geq 3$, then find $p^{-1}(x)$.

Question 6

Core skill

No calculator

Find the inverse of $r(x) = \frac{2x+1}{x-4}$ and list the excluded input of each function.

Section B: Application, comparison, and error analysis

Question 7

Core skill

No calculator

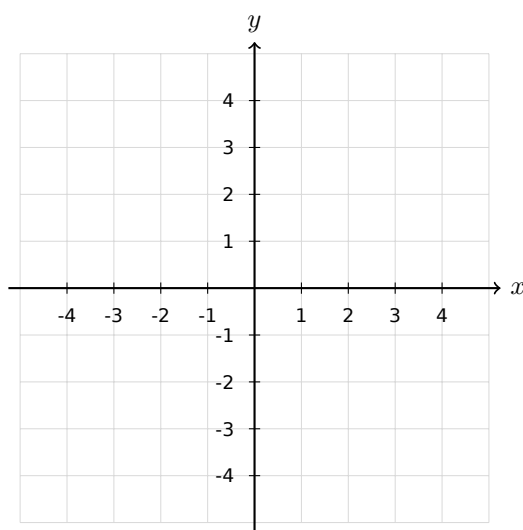
If $f(-3) = 8$, $f(1) = 0$, and $f(5) = -2$, give three values of f^{-1} .

Question 8

Graphing

No calculator

A graph of f contains points $(-1, 4)$, $(2, 7)$, $(6, 9)$. Plot the corresponding inverse points and describe the reflection.



Question 9

Error analysis

No calculator

A student swaps x and y in $y = 2x - 5$ and stops at $x = 2y - 5$. Explain why this is incomplete.

Question 10

Context

No calculator

A taxi model $C(d) = 8 + 2.4d$ gives cost in QAR from distance d in km. Find C^{-1} and interpret $C^{-1}(32)$.

Question 11

Core skill

No calculator

Let $f(x) = \sqrt{x-1} + 3$. Find the inverse and verify the domain-range swap.

Question 12

Parameter reasoning

No calculator

For what values of a is $f(x) = ax + 4$ invertible on $\mathbb{R}$? Give the inverse.

Section C: AP-style multiple choice

MCQ 1

No calculator

Which function is not one-to-one on all real numbers?

- A. $3x - 1$
- B. $x^3 + 2$
- C. 2^x
- D. $(x - 4)^2$

MCQ 2

No calculator

If $f(6) = -3$, which statement is true?

- A. $f^{-1}(6) = -3$
- B. $f^{-1}(-3) = 6$
- C. $f(-3) = 6$
- D. $f^{-1}(3) = -6$

MCQ 3

No calculator

On $x \geq 0$, the inverse of $f(x) = x^2 + 5$ is:

- A. $\sqrt{x} + 5$
- B. $\sqrt{x - 5}$
- C. $-\sqrt{x - 5}$
- D. $x^2 - 5$

MCQ 4

No calculator

The graphs of a function and its inverse are symmetric about:

- A. The x -axis
- B. The y -axis
- C. $y = x$
- D. $y = -x$

Section D: Free-response reasoning

FRQ 1 - Restricted quadratic inverse

No calculator

6 points

Let $f(x) = (x + 2)^2 - 7$ with domain $x \geq -2$. (a) Explain why the domain restriction is needed. (b) Find $f^{-1}(x)$ and its domain. (c) Verify $f^{-1}(f(x)) = x$ on the stated domain. (d) State the reflection relationship between the two graphs.

FRQ 2 - Rational inverse and context

No calculator

6 points

A calibration function is $R(s) = \frac{5s + 12}{s + 2}$ for $s \neq -2$. (a) Find R^{-1} . (b) State the excluded input and output values of R . (c) Verify one composition algebraically. (d) Explain what $R^{-1}(4)$ means if s is a sensor setting and R is the displayed reading.

Topic 2.9 - Logarithmic Expressions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Evaluate logarithmic expressions by translating between logarithmic and exponential forms.
- Approximate common, natural, and arbitrary-base logarithms with technology.
- Interpret logarithms as exponents in mathematical, scientific, and contextual scales.

Section A: Foundation and representation

Question 1

Core skill**No calculator**

Rewrite each statement in exponential form: (a) $\log_2 32 = 5$; (b) $\ln e^{-3} = -3$; (c) $\log_{1/3} 9 = -2$.

Question 2

Core skill**No calculator**

Rewrite each statement in logarithmic form: (a) $7^2 = 49$; (b) $10^{-4} = 0.0001$; (c) $e^{1/2} = \sqrt{e}$.

Question 3

Core skill

No calculator

Evaluate exactly: $\log_5 625$, $\log_8 2$, and $\log_{1/4} 64$.

Question 4

Core skill

No calculator

Evaluate $\log_{16}(1/8)$ exactly.

Question 5

Core skill

No calculator

Solve $\log_b 64 = 2$ for the positive base b .

Question 6

Core skill

No calculator

Solve $\log_3(x - 4) = 5$.

Section B: Application, comparison, and error analysis

Question 7

Core skill

No calculator

State the domain of $\log_2(x^2 - 9)$.

Question 8

Core skill

Calculator permitted

Approximate $\log_5 73$ to four decimal places using change of base.

Question 9

Reasoning

No calculator

Order $\log_2 12$, $\log_3 12$, and $\log_{10} 12$ from least to greatest without calculating decimals.

Question 10

Context

No calculator

The decibel model is $D = 10 \log(I/I_0)$. How many decibels correspond to intensity $1000I_0$?

Question 11

Error analysis

No calculator

A student claims $\log_2(-8) = -3$ because $2^{-3} = 1/8$. Diagnose the error.

Question 12

Scale interpretation

Calculator permitted

A base-10 logarithmic scale increases from 2.4 to 5.1. By what factor does the underlying quantity increase?

Section C: AP-style multiple choice**MCQ 1**

No calculator

What is $\log_4(1/64)$?

- A. -3
- B. -2
- C. 2
- D. 3

MCQ 2

No calculator

Which is equivalent to $\log_7 x = 3$?

- A. $x^7 = 3$
- B. $7^x = 3$
- C. $7^3 = x$
- D. $3^7 = x$

MCQ 3

No calculator

Which expression has domain $x > 5$?

- A. $\log(x - 5)$
- B. $\log(5 - x)$
- C. $\log(x^2 - 25)$
- D. $\log(x + 5)$

MCQ 4

No calculator

If $\log_b 20 = 2.5$, then:

- A. $b^{2.5} = 20$
- B. $20^{2.5} = b$
- C. $b^{20} = 2.5$
- D. $2.5^b = 20$

Section D: Free-response reasoning**FRQ 1 - Exact and approximate logarithms**

Calculator permitted

6 points

(a) Evaluate $\log_{27} 9$ exactly. (b) Solve $\log_4(x + 3) = -2$. (c) Approximate $\log_6 145$ to three decimal places. (d) State the domain of $\ln(2x^2 - 18)$.

FRQ 2 - Logarithmic scale

Calculator permitted

6 points

The model $S = 20 \log(A/A_0)$ compares an amplitude A with reference amplitude A_0 . (a) Find S when $A = 100A_0$. (b) Solve for A/A_0 when $S = 54$. (c) Compare the underlying amplitude ratios for scale values 30 and 50. (d) Explain why adding 20 to the scale multiplies rather than adds to the amplitude ratio.

Topic 2.10 - Inverses of Exponential Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Explain logarithms as inverse functions of exponentials.
- Connect inverse pairs through compositions, points, graphs, domains, ranges, and asymptotes.
- Construct inverses of basic and transformed exponential functions.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Simplify: (a) $\log_7(7^{x+4})$; (b) $7^{\log_7(2x-5)}$ with its restriction.

Question 2

Core skill

No calculator

If $f(x) = 3^x$, list three corresponding points on f^{-1} using $x = -1, 0, 2$ on f .

Question 3

Core skill

No calculator

Find the inverse of $f(x) = 5^x + 8$ and state its domain, range, and vertical asymptote.

Question 4

Core skill

No calculator

Find the inverse of $g(x) = 2(4)^{x+3} - 7$.

Question 5

Core skill

No calculator

Find the inverse of $h(x) = 9 - 6(2)^x$ and state the inverse domain.

Question 6

Core skill

No calculator

Verify that $f(x) = 2^x - 1$ and $g(x) = \log_2(x + 1)$ are inverse functions.

Section B: Application, comparison, and error analysis

Question 7

Context

Calculator permitted

The exponential $P(t) = 500(1.12)^t$ maps time to population. Find the inverse and interpret $P^{-1}(1000)$.

Question 8

Structure

No calculator

An exponential has horizontal asymptote $y = -4$ and range $(-4, \infty)$. What are the inverse domain and vertical asymptote?

Question 9

Error analysis

No calculator

A student claims the inverse of $2^x + 5$ is $2^{-x} - 5$. Explain the error.

Question 10

Parameter reasoning

No calculator

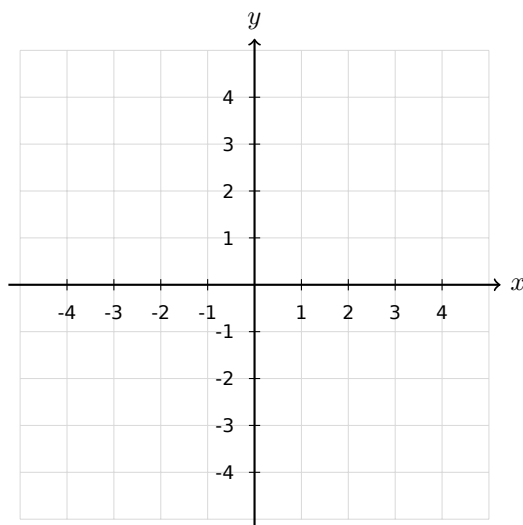
Find a and k if an exponential $f(x) = a(3)^x + k$ has inverse domain $(2, \infty)$ and $f(0) = 7$, assuming $a > 0$.

Question 11

Graphing

No calculator

Sketch $f(x) = 2^x + 1$ and its inverse on the same axes. Label both asymptotes and two reflected point pairs.



Question 12

Inverse reasoning

No calculator

Solve $f^{-1}(x) = 3$ without first writing a formula if $f(x) = 4(2)^{x-1} + 6$.

Section C: AP-style multiple choice

MCQ 1

No calculator

The inverse of $f(x) = 3^x + 4$ is:

- A. $3^{-x} - 4$
- B. $\log_3(x) + 4$
- C. $\log_3(x - 4)$
- D. $\log_3(4 - x)$

MCQ 2

No calculator

If the horizontal asymptote of an exponential is $y = -2$, the vertical asymptote of its inverse is:

- A. $x = -2$
- B. $y = -2$
- C. $x = 2$
- D. $y = 2$

MCQ 3

No calculator

Which identity is valid for $x > 0$?

- A. $\log_b x = b^x$
- B. $b^{\log_b x} = x$
- C. $\log_b(b + x) = x$
- D. $b \log_b x = x$

MCQ 4

No calculator

If $(2, 17)$ lies on f , which point lies on f^{-1} ?

- A. $(-2, -17)$
- B. $(17, 2)$
- C. $(-17, -2)$
- D. $(2, 17)$ only

Section D: Free-response reasoning

FRQ 1 - Transformed inverse

No calculator

6 points

Let $f(x) = 8(3)^{x-2} - 5$. (a) Find $f^{-1}(x)$. (b) State the domain, range, and vertical asymptote of f^{-1} . (c) Verify $f^{-1}(f(x)) = x$. (d) Identify the inverse point corresponding to the anchor point of f .

FRQ 2 - Inverse model in context

Calculator permitted

6 points

A fund is modeled by $B(t) = 12000(1.045)^t$, where t is years and B is QAR. (a) Find B^{-1} and state its domain. (b) Interpret the inverse function. (c) Estimate when the balance reaches 18000 QAR. (d) Explain one limitation of interpreting $B^{-1}(y)$ for every $y > 0$.

Topic 2.11 - Logarithmic Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Describe logarithmic domain, range, asymptotes, monotonicity, concavity, and end behavior.
- Connect equal multiplicative changes in input to equal additive changes in output.
- Analyze and construct transformed logarithmic functions with explicit restrictions.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Analyze $f(x) = \log_5 x$: state domain, range, intercepts, asymptote, monotonicity, concavity, and end behavior.

Question 2

Core skill

No calculator

For $g(x) = -2 \log_3(x + 1) + 4$, state the domain, asymptote, anchor point, monotonicity, and concavity.

Question 3

Core skill

No calculator

Find the x -intercept of $h(x) = 3 \log_2(x - 4) - 6$.

Question 4

Core skill

No calculator

Find the domain and asymptote of $p(x) = \ln(2 - 3x) + 5$.

Question 5

Core skill

No calculator

Construct a function of the form $f(x) = a \log_4(x - 1) + k$ that passes through $(2, 3)$ and $(5, 7)$.

Question 6

Comparison

No calculator

Compare $\log_2 x$ and $\log_8 x$ for $x > 1$ and for $0 < x < 1$.

Section B: Application, comparison, and error analysis

Question 7

Core skill

No calculator

For $f(x) = 5 \log_3 x + 2$, find $f(27x) - f(x)$.

Question 8

Multiplicative change

No calculator

A logarithmic model $L(x) = 12 + 7 \log_2 x$ is defined for $x > 0$. By what factor must x be multiplied to increase L by 21?

Question 9

Error analysis

No calculator

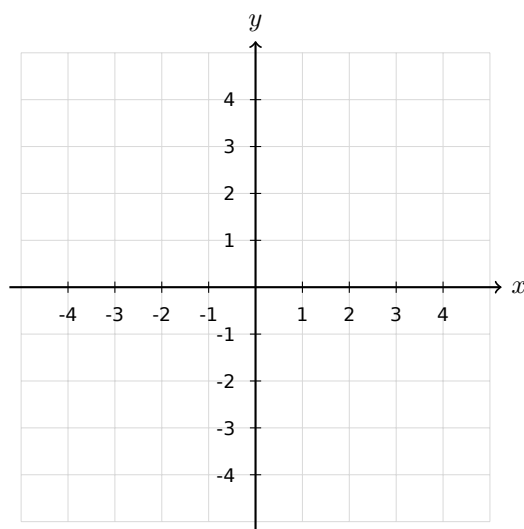
A student states the domain of $\log(x^2 - 4)$ is $x \neq \pm 2$. Correct the error.

Question 10

Graphing

No calculator

Sketch $q(x) = \log_2(x + 2) - 1$, labeling the asymptote, anchor point, and x -intercept.



Question 11

Parameter reasoning

No calculator

Determine all values of a for which $f(x) = a \ln x$ is decreasing and concave up.

Question 12

Context

No calculator

A model is $S(x) = 30 + 8 \log_{10}(x/5)$ for $x > 0$. Interpret the change in S when x is multiplied by 100.

Section C: AP-style multiple choice

MCQ 1

No calculator

The vertical asymptote of $f(x) = \log_3(x - 7) + 2$ is:

- A. $x = -7$
- B. $x = 7$
- C. $y = 2$
- D. $y = 7$

MCQ 2

No calculator

Which function is decreasing and concave up on its domain?

- A. $\log_2 x$
- B. $-\log_2 x$
- C. 2^x
- D. -2^x

MCQ 3

No calculator

The anchor point of $4\log_5(x + 3) - 2$ is:

- A. $(-3, -2)$
- B. $(-2, -2)$
- C. $(1, 2)$
- D. $(3, -2)$

MCQ 4

No calculator

For $f(x) = 3\log_2 x$, what is $f(4x) - f(x)$?

- A. 2
- B. 3
- C. 6
- D. 12

Section D: Free-response reasoning

FRQ 1 - Complete logarithmic analysis

No calculator

6 points

Let $f(x) = -4\log_3(x-2)+6$. (a) State domain, range, vertical asymptote, and anchor point. (b) Find the x -intercept. (c) Describe monotonicity, concavity, and end behavior. (d) Find $f(9(x-2)+2)-f(x)$ for $x > 2$.

FRQ 2 - Construct and interpret

No calculator

6 points

A logarithmic model has vertical asymptote $x = 5$. It passes through $(6, 2)$ and $(14, 8)$, and uses base 3. (a) Construct the model in the form $a\log_3(x-5)+k$. (b) State its domain and monotonicity. (c) Find the input at which the model output is 5. (d) Explain how the output changes when $x - 5$ is multiplied by 27.

Topic 2.12 - Logarithmic Function Manipulation

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Expand and condense logarithmic expressions using product, quotient, and power properties.
- Use change of base and natural logarithms to evaluate and compare logarithms.
- Explain algebraic and graphical equivalences while preserving domain restrictions.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Expand $\log_5(125x^4y)$ for $x > 0, y > 0$.

Question 2

Core skill

No calculator

Expand $\ln\left(\frac{e^3(x+1)^2}{\sqrt{x}}\right)$ for $x > 0$.

Question 3

Core skill

No calculator

Expand $\log_3 \left(\frac{(x-2)^6}{9(x+4)} \right)$ and state the original domain.

Question 4

Core skill

No calculator

Condense $3\log_2 x + \log_2(x-1) - 2\log_2(x+3)$ and state the domain.

Question 5

Core skill

No calculator

Condense $\frac{1}{2} \ln a + \frac{3}{2} \ln b - \ln c$ for positive a, b, c .

Question 6

Core skill

Calculator permitted

Use change of base to approximate $\log_{12} 500$ to four decimal places.

Section B: Application, comparison, and error analysis

Question 7

Proof

No calculator

Prove that $\log_{1/3} x = -\log_3 x$ for $x > 0$.

Question 8

Domain reasoning

No calculator

Show that $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ are equivalent only on their shared domain. Compare their domains.

Question 9

Graph equivalence

No calculator

Describe the graph relationship between $f(x) = \log_5(25x)$ and $g(x) = \log_5 x$.

Question 10

Error analysis

No calculator

A student condenses $\ln(x - 2) + \ln(x + 2)$ to $\ln(x^2 - 4)$ and changes the domain to $x < -2$ or $x > 2$. Explain the domain error.

Question 11

Core skill

No calculator

Solve for k if $\log_2(8x) = \log_2 x + k$ for all $x > 0$.

Question 12

Core skill

No calculator

Rewrite $4\log_3 x - 2\log_3 y + \log_3 9$ as a single logarithm.

Section C: AP-style multiple choice**MCQ 1**

No calculator

Which is equivalent to $\log_b(MN^2)$ for positive M, N ?

- A. $\log_b M + 2\log_b N$
- B. $\log_b M \cdot 2\log_b N$
- C. $\log_b(M + N^2)$
- D. $2\log_b(MN)$

MCQ 2

No calculator

Which expression condenses to $\log(x^3/y)$?

- A. $3\log x - \log y$
- B. $\log(3x) - \log y$
- C. $\log x - 3\log y$
- D. $\log(x/y^3)$

MCQ 3

No calculator

Which equality is false in general?

- A. $\log_b(xy) = \log_b x + \log_b y$
- B. $\log_b(x/y) = \log_b x - \log_b y$
- C. $\log_b(x + y) = \log_b x + \log_b y$
- D. $\log_b(x^p) = p \log_b x$ on the shared domain

MCQ 4

No calculator

$\log_{25} x$ is equal to:

- A. $2 \log_5 x$
- B. $\frac{1}{2} \log_5 x$
- C. $\log_5 x + 2$
- D. $\log_5(x/2)$

Section D: Free-response reasoning

FRQ 1 - Expand, condense, and preserve domain

No calculator

6 points

Let $E(x) = 2\ln(x - 1) + \frac{1}{2}\ln(x + 3) - \ln 4$. (a) State the domain of E . (b) Condense E into one logarithm. (c) Expand $\ln\left(\frac{(x - 1)^2\sqrt{x + 3}}{4}\right)$ and explain why the domain must remain the same as in part (a). (d) Evaluate $E(5)$ exactly.

FRQ 2 - Change of base and graphical meaning

No calculator

6 points

Define $f(x) = \log_2(16x)$ and $g(x) = \log_8 x$ for $x > 0$. (a) Rewrite each function using $\log_2 x$. (b) Describe the graph transformation from $y = \log_2 x$ to each function. (c) Solve $f(x) = g(x)$. (d) Find the common output.

Topic 2.13 - Exponential and Logarithmic Equations and Inequalities

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Solve exponential equations and inequalities exactly or approximately.
- Solve logarithmic equations and inequalities while enforcing domain restrictions.
- Interpret solutions and thresholds in transformed functions and contextual models.

Section A: Foundation and representation

Question 1

Core skill**No calculator**

Solve $8^{x+1} = 4^{2x-1}$.

Question 2

Core skill**Calculator permitted**

Solve $7^{2x-3} = 20$ exactly and to three decimal places.

Question 3

Core skill

No calculator

Solve $4^x - 5(2^x) + 4 = 0$.

Question 4

Core skill

No calculator

Solve $3^{x+1} = 5^{x-2}$ exactly.

Question 5

Core skill

No calculator

Solve $\log_3(x^2 - 4) = 2$.

Question 6

Core skill

No calculator

Solve $\log x + \log(x - 9) = 1$.

Section B: Application, comparison, and error analysis

Question 7

Core skill

No calculator

Solve $\ln(x + 4) = 2 \ln x$.

Question 8

Core skill

No calculator

Solve $2^{x-3} < 1/16$.

Question 9

Core skill

No calculator

Solve $\log_{1/3}(2x - 1) \geq -2$.

Question 10

Error analysis

No calculator

A student solves $\ln(x - 2) + \ln(x + 2) = \ln 5$ and keeps both roots of $x^2 - 4 = 5$. Correct the solution.

Question 11

Context threshold

Calculator permitted

A medication model is $A(t) = 180(0.78)^t$. Find when the amount first falls below 25 mg, with t measured in whole hours.

Question 12

Inequality

No calculator

Solve $5 - 2\log_3(x - 1) \leq 1$.

Section C: AP-style multiple choice**MCQ 1**

No calculator

The exact solution of $3^x = 11$ is:

- A. $x = 11/3$
- B. $x = \log_3 11$
- C. $x = 3^{11}$
- D. $x = \ln 3 / \ln 11$

MCQ 2

No calculator

Which candidate must be rejected when solving $\log(x - 2) + \log(x + 2) = 1$?

- A. $x = \sqrt{14}$
- B. $x = -\sqrt{14}$
- C. Both
- D. Neither

MCQ 3

No calculator

The solution of $(1/4)^x > 16$ is:

- A. $x > -2$
- B. $x < -2$
- C. $x > 2$
- D. $x < 2$

MCQ 4

No calculator

Which step is essential after solving a logarithmic equation?

- A. Differentiate the result
- B. Check candidates in the original domain
- C. Convert every value to a decimal
- D. Graph only the left side

Section D: Free-response reasoning**FRQ 1 - Mixed exact solutions**

No calculator

6 points

(a) Solve $9^x - 10(3^x) + 9 = 0$. (b) Solve $\log_2(x - 1) + \log_2(x - 7) = 4$. (c) State and apply all domain restrictions. (d) Verify the valid logarithmic solution.

FRQ 2 - Thresholds and inequalities

Calculator permitted

6 points

A model is $P(t) = 900(1.13)^t$, where t is measured in whole years. (a) Solve $P(t) \geq 2500$ over real t . (b) State the first whole year satisfying the inequality. (c) Solve $900(1.13)^t < 700$ and interpret the result. (d) Explain how the answers would change if the factor were 0.87 instead of 1.13.

Topic 2.14 - Logarithmic Function Context and Data Modeling

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Construct logarithmic models from proportional input change, two points, transformations, or regression output.
- Use logarithmic models to predict outputs and solve for inputs.
- Interpret domains, units, scale meaning, diminishing change, and model limitations.

Section A: Foundation and representation

Question 1

Core skill**No calculator**

A response is 18 at $x = 1$ and increases by 7 whenever x is multiplied by 4. Write a model.

Question 2

Core skill**No calculator**

Construct $f(x) = a \log_3 x + k$ through $(1, -2)$ and $(27, 10)$.

Question 3

Core skill

No calculator

A model has vertical asymptote $x = 5$, anchor point $(6, 11)$, and output increases by 8 whenever $x - 5$ is multiplied by 2. Write the model.

Question 4

Core skill

No calculator

For $S(x) = 24 + 6 \ln x$, find $S(1)$, $S(e^2)$, and the change from x to e^3x .

Question 5

Core skill

Calculator permitted

Solve $40 + 5 \log_{10} x = 57$ for x .

Question 6

Interpretation

Calculator permitted

A fitted model is $y = 72 + 9.5 \ln x$. Interpret the coefficient 9.5 in terms of multiplying the input by e and by 10.

Section B: Application, comparison, and error analysis

Question 7

Context

No calculator

A learning model is $L(h) = 45 + 12 \log_2(h + 1)$, where $h \geq 0$ is practice hours. Find the predicted gain when practice increases from 3 to 7 hours.

Question 8

Scale reasoning

No calculator

A model $D(x) = 20 \log(x/x_0)$ uses a positive reference x_0 . Show that multiplying x by 100 increases D by 40.

Question 9

Error analysis

No calculator

A student fits $y = a + b \ln x$ to data containing $x = 0$. Explain the problem and suggest one possible remedy.

Question 10

Comparison

Calculator permitted

Compare $f(x) = 5 \ln x$ and $g(x) = 2 \log_2 x$ by rewriting both with natural logarithms. Which grows faster as a function of $\ln x$?

Question 11

Limitation

No calculator

A logarithmic model predicts an output of -40 near its positive domain boundary, but the measured quantity cannot be negative. What should be concluded?

Question 12

Construction

No calculator

Construct a logarithmic model that passes through $(3, 4)$ and $(11, 16)$ and has vertical asymptote $x = 2$, using base 3.

Section C: AP-style multiple choice**MCQ 1**

No calculator

A model increases by 5 whenever the input is multiplied by 2. Which form is consistent?

- A. $5x + 2$
- B. $5 \log_2 x + k$
- C. $2^x + 5$
- D. $k(5)^x$

MCQ 2

No calculator

For $f(x) = a \ln x + k$, multiplying x by e^3 changes f by:

- A. 3
- B. $a + 3$
- C. $3a$
- D. e^{3a}

MCQ 3

No calculator

The domain of $7 + 4\log(x - 6)$ is:

- A. $x < 6$
- B. $x > 6$
- C. $x \neq 6$
- D. All real x

MCQ 4

No calculator

Which statement is a valid model limitation?

- A. Logarithmic functions can never model data.
- B. A high fit statistic proves causation.
- C. The formula may be undefined or unrealistic near the domain boundary.
- D. Logarithmic growth becomes negative for all large inputs.

Section D: Free-response reasoning**FRQ 1 - Construct a logarithmic model**

No calculator

6 points

A response model has vertical asymptote $x = 2$. It passes through $(3, 10)$ and $(11, 22)$, and uses base 3. (a) Construct the model. (b) State its domain and interpret the anchor point. (c) Find the input that gives output 16. (d) Describe the output change when $x - 2$ is multiplied by 27.

FRQ 2 - Evaluate a fitted model

Calculator permitted

6 points

A fitted model is $S(x) = 38 + 7.2 \ln x$ for $1 \leq x \leq 50$. (a) Interpret $S(1)$ and the coefficient 7.2. (b) Predict $S(10)$. (c) Find the input that gives $S = 60$. (d) Explain why predictions at $x = 0.1$ and $x = 5000$ require caution.

Topic 2.15 - Semi-log Plots

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Interpret semi-log plots with a logarithmically scaled output axis.
- Use linearity in x versus $\log y$ to identify and construct exponential models.
- Explain how slope, intercept, base, shifts, and limitations affect linearization.

Section A: Foundation and representation

Question 1

Core skill

Calculator permitted

Linearize $y = 18(1.25)^x$ using common logarithms. Identify the slope and intercept.

Question 2

Core skill

Calculator permitted

A line on a plot of x versus $\ln y$ is $\ln y = 2.4 + 0.18x$. Write the exponential model.

Question 3

Core skill

Calculator permitted

A line on a base-10 semi-log plot is $\log y = -0.25x + 3$. Find the initial value and one-unit factor.

Question 4

Core skill

No calculator

Two points on a $\ln y$ versus x line are $(2, 3.1)$ and $(7, 4.6)$. Find the exponential model $y = ab^x$.

Question 5

Core skill

Calculator permitted

On a base-10 semi-log line, increasing x by 5 raises $\log y$ by 1.2. Find the factor for a one-unit increase and for a five-unit increase.

Question 6

Core skill

No calculator

A semi-log line has slope $\log 0.84$. What percent change occurs per one-unit increase in x ?

Section B: Application, comparison, and error analysis

Question 7

Core skill

Calculator permitted

For the data below, compute $\log_{10} y$ and decide whether an exponential model is plausible.

x	0	1	2	3
y	20	40	80	160

Question 8

Shifted model

No calculator

A shifted model is $P(t) = 50 + 300(0.9)^t$. Describe the correct quantity to plot on a logarithmic scale to obtain a line.

Question 9

Error analysis

No calculator

A student reads equal vertical spacing between 10 and 100 and between 100 and 1000 as equal output differences. Correct the interpretation.

Question 10

Validation

No calculator

A data set looks linear on both an ordinary plot and a semi-log plot over only three points. What should be done before choosing a model?

Question 11

Base reasoning

No calculator

If $\log_2 y = 4 - 0.5x$, write y as an exponential and identify the factor per two-unit increase in x .

Question 12

Limitation

No calculator

A semi-log model fitted on $0 \leq x \leq 8$ is extrapolated to $x = 80$. State two reasons for caution.

Section C: AP-style multiple choice**MCQ 1**

No calculator

If $\log_{10} y = 0.2x + 1$, the exponential factor per unit is:

- A. 0.2
- B. 10
- C. $10^{0.2}$
- D. 1.2

MCQ 2

No calculator

A straight decreasing line on a plot of x versus $\ln y$ suggests:

- A. Linear growth in y
- B. Exponential decay in y
- C. Quadratic growth in y
- D. A logarithmic model for y

MCQ 3

No calculator

For $y = L + ab^x$, which transformation is most likely to linearize the data?

- A. Plot $y - L$ against x on ordinary axes
- B. Plot $\log(y - L)$ against x
- C. Plot $\log x$ against y
- D. Plot $1/y$ against x

MCQ 4

No calculator

Equal vertical distances on a base-10 log axis represent equal:

- A. Differences
- B. Ratios
- C. Slopes
- D. Intercepts

Section D: Free-response reasoning

FRQ 1 - Construct from a semi-log line

Calculator permitted

6 points

A fitted line on a plot of x versus $\log_{10} y$ is $\log y = 1.3 + 0.12x$. (a) Write the exponential model $y = ab^x$. (b) Interpret a and b . (c) Predict y at $x = 10$. (d) Find the input at which the model reaches $y = 500$.

FRQ 2 - Shifted exponential linearization

Calculator permitted

6 points

A process is believed to follow $T(t) = L + Ab^t$. A known limiting value is $L = 18$. Measurements are $T(0) = 82$, $T(2) = 54$, and $T(4) = 38.25$. (a) Compute $T - L$ for each time. (b) Use the transformed values to test whether one exponential factor is plausible. (c) Construct a model. (d) Explain what would happen if $\log T$ were plotted instead of $\log(T - 18)$.
